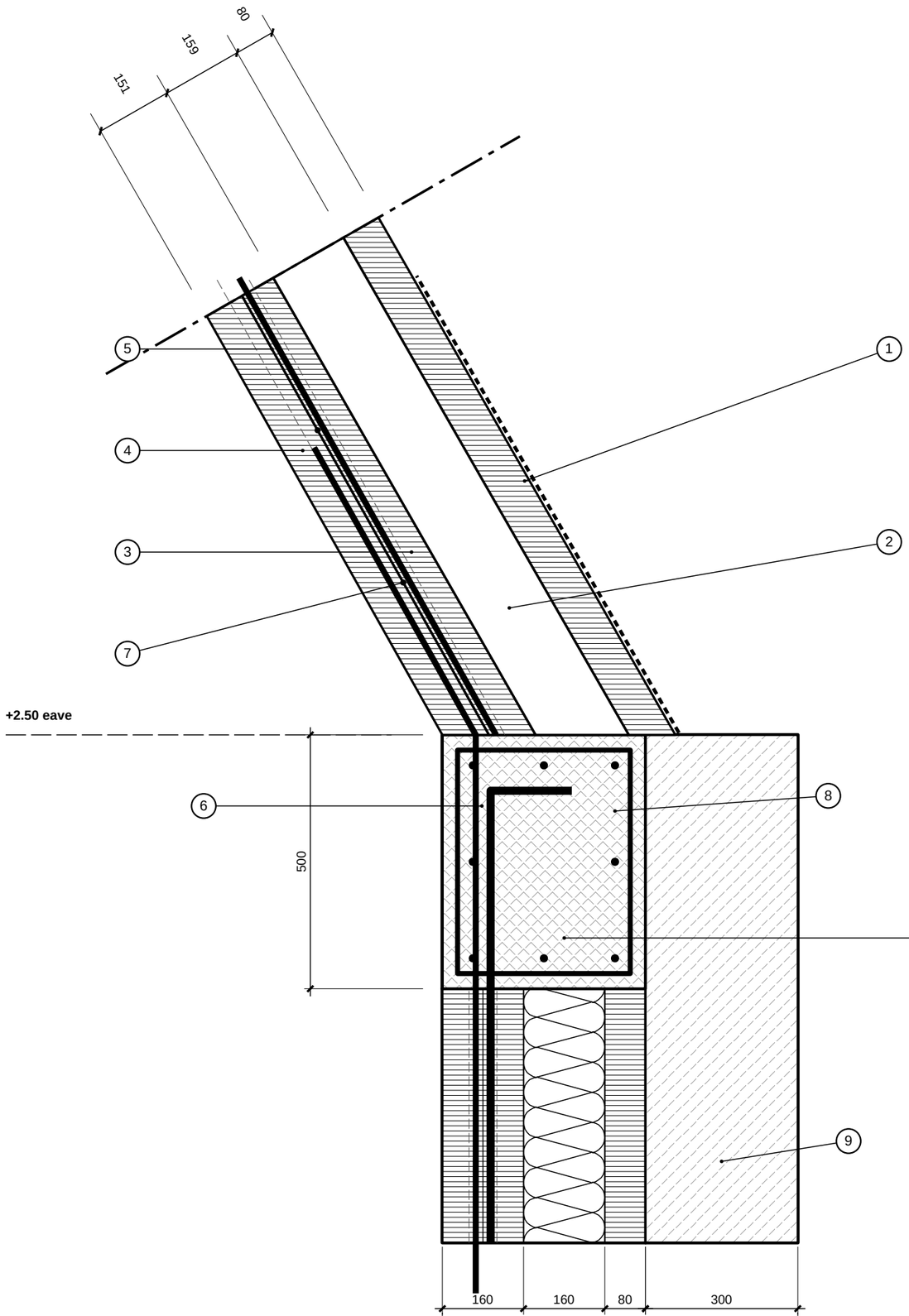
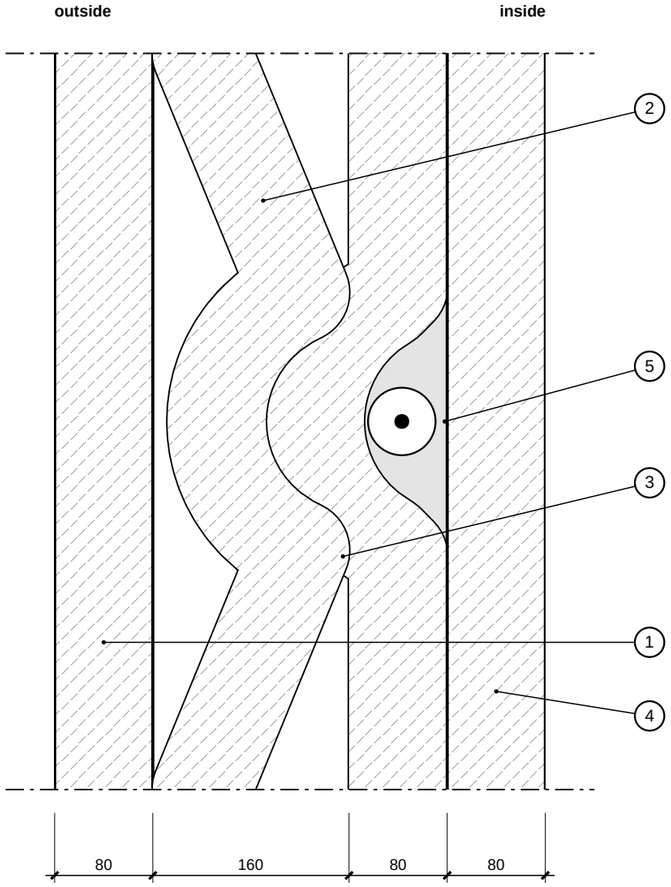


Plan of one roof piece, 1:5

layer pattern, bars every ≈ 600 along the path



Detail D1: eave, roof to wall

at the mid-span buttress, section A-A, 1:10 · piece plan 1:5

1. Roof outer layer 80, printed with the rest of the piece; cladding on the outside face (dashed line).
2. Zig-zag bead 80 in a 160 zone; it touches the outer layer and layer 2 every ≈ 600; the cells between are open air pockets for ventilation.
3. Layer 2, 80: printed bead that curves to leave a pocket for the bar.
4. Inner layer 1, 80, room side. Layers 1 and 2 together are the structural leaf: 160 at the base, 80 at the crown (layer 2 ends).
5. Rebar pocket Ø55 in the leaf, grouted after placing. Roof bars R Ø12 every ≈ 600 along the path, lapped 650 with the knee bars B.
6. Knee bars B Ø12 every ≈ 600, one continuous bar over the eave: 1100 down the wall pocket (laps wall bars A), through the beam, 650 up the roof.
7. Ring bars Ø12 in the pocket zone every 300 (assumed).
8. Ring beam 400 x 500, cast in place: 8 Ø16, closed stirrups Ø10 / 150; wall bars A hook into it. Embedded in the buttress notch.
9. Buttress 400 wide, 700 from the inner face (300 beyond the wall), L-shaped around the ring beam; top flush with the roof base at +2.50.
10. Wall 400 printed, same pattern as the roof; wall bars A Ø16 run from the footing up into the beam and end in a 160 hook (assumed).

● from your sketches
○ assumed by me, to confirm

Notes

Layer pattern from your drawing: from the inside, 80 + 80 printed layers (the second curves round the bar), then a 160 zig-zag zone, then the 80 outer layer; 400 in total.

Wall and roof are 3D printed in 10 mm layers (horizontal joints drawn every 10). Roof thicknesses are square to the slope; the wall is vertical. All rebar is black and drawn at true diameter (Ø16 = 1.6 mm on the sheet); bar sizes and spacings are assumed, to be checked by the structural engineer. Black lines and dots = rebar, white lattice cells = open air, thin horizontal lines = printed layers, single hatch = other concrete, crossed hatch = ring beam cut in the section plane.

0 1 2 3 4 5 dm

Scale 1:10 at A3 · dimensions in mm · draft 1, 2026-10-11
pyramid-roof · hand estimates, not engineered